

Table 11. The prompt for idea integration.

System Message	Now you are a researcher in the field of AI. You are good at selecting the ideas that meet the requirements.
User Message	### Task Description: You will be provided with some ideas you previously generated, and a research background. Your task is to select 5-6 ideas that best address the problems described in the research background (priority) and ideas that are relatively novel and feasible (secondary). ### Information Provided: 1. Ideas: These are the ideas you previously generated based on the research background and several related papers. 2. Research Background: This document describes specific problems and challenges that need to be addressed. ### Approach: Your approach should be systematic: - Step 1: Analyze the research background to understand the specific problems that need solutions. - Step 2: Critically review the ideas, selecting 5-6 ideas that are most effective in solving the problems in the research background (priority) and that are also relatively novel and feasible (secondary). ### Specific Information: I will provide you with specific information now; please use them according to the instructions above: 1. Ideas: {idea} 2. Research Background: {background} ### Format for Your Response: Please ensure that your final ideas include 5-6 entries, whose content has not been modified. Don't generate any explanation and just present the filtered ideas as well as their content in the following format: Idea 1: [The first method idea] Idea 2: [The second method idea] Idea 3: [The third method idea] ...

Table 12. The prompt for idea expansion.

System Message	Now you are a researcher in the field of AI with innovative and pioneering abilities. You are good at transforming a brief scientific idea into a concrete algorithm.
User Message	# Task Description: You are an AI researcher conducting studies in a specific domain. Someone has provided you with a brief scientific idea, and your task is to transform it into a detailed, feasible, and concrete algorithm. If necessary, you may incorporate formulas to elaborate on the algorithm in the Latex format. I will give you an example. The example begins with “# Example 1” and includes a Brief Scientific Idea and its corresponding Detailed Scientific Idea. Then, your task starts with “# Your Task”, containing “Your Brief Scientific Idea”. Your job is to expand Your Brief Scientific Idea into a Detailed Scientific Idea by referring to Example 1. Note that the ideas in Example 1 are unrelated to your idea, so the key focus should be on the relationship between the Brief Scientific Idea and the Detailed Scientific Idea. You should directly start with your response and do not start with a section title like “## Detailed Scientific Idea”. # Example 1 ## Example Brief Scientific Idea {Example brief idea} ## Example Detailed Scientific Idea {Example detailed idea} # Your Task ## Your Research Background {background} ## Your Brief Scientific Idea {brief_idea}